Phần 2:

- Bài 1:

docker-compose.yaml U

22678601_PhanTanDuy_Tuan05 > Bai01 > docker-compose.yaml

```yaml
version: "3.8" # Phiên bản cấu hình Docker Compose

services:
  my-nginx: # Tên dịch vụ bạn tự đặt
    image: nginx:latest # Sử dụng bản Nginx mới nhất từ Docker Hub
    ports:
      - "8080:80" # Map cổng: <Máy_Host>:<Container>
```

C:\Windows\System32\cmd.exe

```
Microsoft Windows [Version 10.0.19045.6216]
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T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai01>docker compose up -d
[+] Running 8/8
 ⠿ my-nginx 7 layers [⠿⠿⠿⠿⠿⠿⠿]      0B/0B      Pulled                                                    11.1s
   ⠿ ec781dee3f47 Pull complete                                                                            4.3s
   ⠿ bb3d0aa29654 Pull complete                                                                            4.4s
   ⠿ 510ddf6557d6 Pull complete                                                                            1.3s
   ⠿ cde7a05ae428 Pull complete                                                                            2.5s
   ⠿ 587e3d84dbb5 Pull complete                                                                            3.7s
   ⠿ 3189680c601f Pull complete                                                                            4.9s
   ⠿ 5e815e07e569 Pull complete                                                                            7.2s
[+] Building 0.0s (0/0)                                                                      docker:default
[+] Running 2/2
 ⠿ Network bai01_default       Created                                                                     0.0s
 ⠿ Container bai01-my-nginx-1  Started                                                                     0.1s

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai01>
```
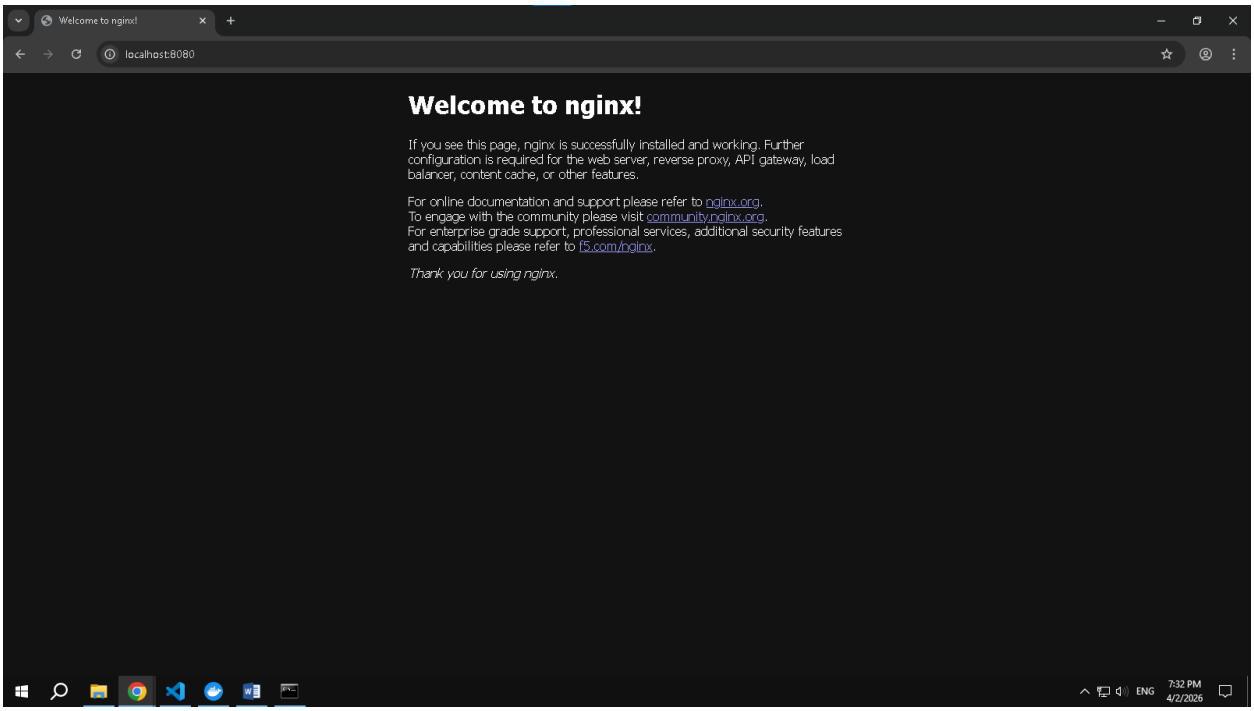

```
T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai01>docker compose ps
NAME               IMAGE          COMMAND                                          SERVICE    CREATED          STATUS          PORTS
bai01-my-nginx-1   nginx:latest   "/docker-entrypoint.sh nginx -g 'daemon off;'"   my-nginx   31 seconds ago   Up 30 seconds   0.0.0.0:8080->80/tcp
```



- Bài 2:

docker-compose.yaml U

22678601_PhanTanDuy_Tuan05 > Bai02 > docker-compose.yaml

```yaml
version: "3.8"

services:
  db:
    image: mysql:8.0
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: mydb
      MYSQL_USER: user
      MYSQL_PASSWORD: password
    ports:
      - "3306:3306"
```

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6216]
(c) Microsoft Corporation. All rights reserved.

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai02>docker compose up -d
[+] Running 12/12
 ⠿ db 11 layers [⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿]      0B/0B      Pulled                                                          25.3s
   ⠿ d4c7048d1cf1 Pull complete                                                                                    14.0s
   ⠿ 6375ec02106f Pull complete                                                                                     1.2s
   ⠿ 771773a1fe7a Pull complete                                                                                     1.4s
   ⠿ c3b8cfb10b41 Pull complete                                                                                     2.9s
   ⠿ 1378718ee973 Pull complete                                                                                     2.6s
   ⠿ 5915b0e5580c Pull complete                                                                                     3.8s
   ⠿ 60c3210ab1f6 Pull complete                                                                                     5.8s
   ⠿ 440208b3f29f Pull complete                                                                                     5.3s
   ⠿ 10ee8ef3b9fc Pull complete                                                                                     9.4s
   ⠿ cd8097e7c26b Pull complete                                                                                     7.0s
   ⠿ 0940ecf5daec Pull complete                                                                                     8.2s
[+] Building 0.0s (0/0)                                                                                  docker:default
[+] Running 2/2
 ⠿ Network bai02_default  Created                                                                                   0.0s
 ⠿ Container bai02-db-1   Started                                                                                   0.5s

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai02>
```

```
T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai02>docker compose ps
NAME          IMAGE       COMMAND                       SERVICE   CREATED         STATUS          PORTS
bai02-db-1    mysql:8.0   "docker-entrypoint.sh mysqld"   db        8 minutes ago   Up 8 minutes    0.0.0.0:3306->3306/tcp, 33060/tcp
```

- Bài 3:

docker-compose.yaml U

22678601_PhanTanDuy_Tuan05 > Bai03 > docker-compose.yaml

```yaml
version: "3.8"

services:
  db:
    image: mysql:8.0
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: mydb
      MYSQL_USER: user
      MYSQL_PASSWORD: password
    ports:
      - "3306:3306"
    volumes:
      - db_data:/var/lib/mysql

  phpmyadmin:
    image: phpmyadmin:latest
    restart: always
    ports:
      - "8081:80"
    environment:
      PMA_HOST: db
      MYSQL_ROOT_PASSWORD: rootpassword

volumes:
  db_data:
```

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.6216]
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T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai03>docker compose up -d
[+] Running 22/22
 ⠿ phpmyadmin 21 layers [⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿]      0B/0B      Pulled                                          15.5s
   ⠿ ec781dee3f47 Already exists                                                                   0.0s
   ⠿ 492c65e7c8ac Pull complete                                                                    1.3s
   ⠿ 60db4f438f38 Pull complete                                                                    5.8s
   ⠿ 7d5440bdc2a2 Pull complete                                                                    1.2s
   ⠿ f0ff10d8b131 Pull complete                                                                    3.6s
   ⠿ fa79359a5906 Pull complete                                                                    2.5s
   ⠿ 0e62964b059f Pull complete                                                                    3.8s
   ⠿ 9ab4f7b41339 Pull complete                                                                    5.6s
   ⠿ a6418fb4b014 Pull complete                                                                    5.0s
   ⠿ b879b42dc415 Pull complete                                                                    6.7s
   ⠿ 7ddb4115d2ee Pull complete                                                                    6.8s
   ⠿ 41d15dbe17b9 Pull complete                                                                    7.0s
   ⠿ 48a25387913f Pull complete                                                                    7.9s
   ⠿ 5a5870c8f47c Pull complete                                                                    8.0s
   ⠿ 4f4fb700ef54 Pull complete                                                                    8.2s
   ⠿ 3e4fc790b65c Pull complete                                                                    9.5s
   ⠿ 2aeefa54aa35 Pull complete                                                                    9.2s
   ⠿ 3e7b3cdf1575 Pull complete                                                                   10.2s
   ⠿ d24a5f7cb1a6 Pull complete                                                                   10.5s
   ⠿ 833ee3620b11 Pull complete                                                                   10.7s
   ⠿ f13a2796fece Pull complete                                                                   11.4s
[+] Building 0.0s (0/0)                                                                  docker:default
[+] Running 4/4
 ⠿ Network ba03_default         Created                                                           0.0s
 ⠿ Volume "ba03_db_data"        Created                                                           0.0s
 ⠿ Container ba03-phpmyadmin-1  Started                                                           1.2s
 ⠿ Container ba03-db-1          Started                                                           1.2s

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai03>docker compose ps
NAME                 IMAGE              COMMAND                  SERVICE      CREATED          STATUS         PORTS
ba03-db-1            mysql:8.0          "docker-entrypoint.sh mysqld"   db           10 seconds ago   Up 8 seconds   0.0.0.0:3306->3306/tcp, 33060/tcp
ba03-phpmyadmin-1    phpmyadmin:latest  "/docker-entrypoint.sh apache2-foreground"   phpmyadmin   10 seconds ago   Up 8 seconds   0.0.0.0:8081->80/tcp

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai03>
```
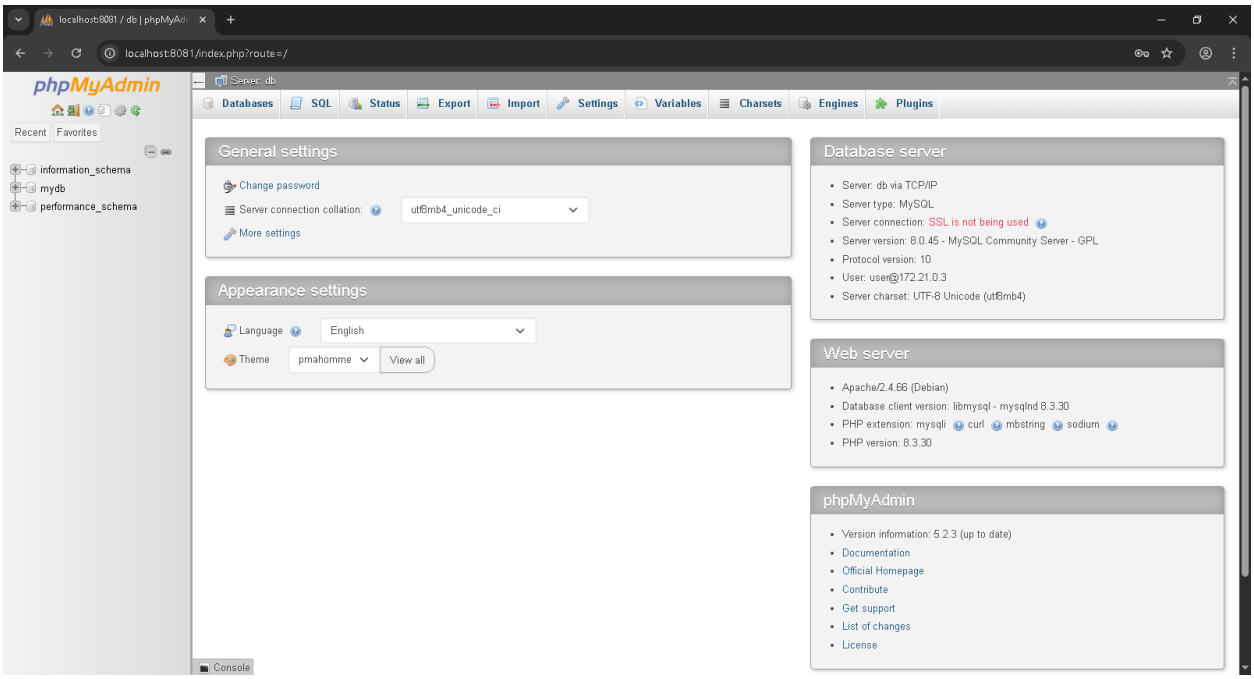

Login với user và password



- Bài 4:

{} package.json U | JS app.js U × | Dockerfile U | docker-compose.yaml U

22678601_PhanTanDuy_Tuan05 > Bai04 > JS app.js > app.get('/') callback

```js
const express = require('express');
const app = express();
const PORT = 3000;

app.get('/', (req, res) => {
  res.send('Chào mừng bạn đến với ứng dụng Node.js chạy trong Docker!');
});

app.listen(PORT, () => {
  console.log(`Server đang chạy tại http://localhost:${PORT}`);
});
```

{} package.json U | JS app.js U | Dockerfile U × | docker-compose.yaml U

22678601_PhanTanDuy_Tuan05 > Bai04 > Dockerfile

```dockerfile
FROM node:18-alpine
WORKDIR /app
COPY package.json .
RUN npm install
COPY . .
CMD ["node", "app.js"]
EXPOSE 3000
```

{} package.json U | JS app.js U | Dockerfile U | docker-compose.yaml U ×

22678601_PhanTanDuy_Tuan05 > Bai04 > docker-compose.yaml

```yaml
version: "3.8"

services:
  webapp:
    build: .
    ports:
      - "5000:3000"
    volumes:
      - .:/app
    environment:
      - NODE_ENV=development
```

```
C:\Windows\System32\cmd.exe

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai04>docker compose up -d --build
[+] Building 10.6s (10/10) FINISHED                                                                  docker:default
 => [webapp internal] load .dockerignore                                                                        0.0s
 => => transferring context: 2B                                                                                 0.0s
 => [webapp internal] load build definition from Dockerfile                                                     0.0s
 => => transferring dockerfile: 155B                                                                            0.0s
 => [webapp internal] load metadata for docker.io/library/node:18-alpine                                        3.2s
 => [webapp 1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb  4.4s
 => => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8c  0.0s
 => => sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f0d26702d6e3 1.26MB / 1.26MB                  1.4s
 => => sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 7.67kB / 7.67kB                  0.0s
 => => sha256:929b04d7c782f04f615cf785488fed452b6569f87c73ff666ad553a7554f0006 1.72kB / 1.72kB                  0.0s
 => => sha256:ee77c6cd7c1886ecc802ad6cedef3a8ec1ea27d1fb96162bf03dd3710839b8da 6.18kB / 6.18kB                  0.0s
 => => sha256:f18232174bc91741fdf3da96d8501109210 1a032a93a388b79e99e69c2d5c870 3.64MB / 3.64MB                  1.0s
 => => sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4efea161b2b5a3d0e 40.01MB / 40.01MB                 3.4s
 => => extracting sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870                       0.1s
 => => sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549 446B / 446B                      1.6s
 => => extracting sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4efea161b2b5a3d0e                       0.8s
 => => extracting sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430bd47f0d26702d6e3                       0.0s
 => => extracting sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9ea70b80df5a2e0549                       0.0s
 => [webapp internal] load build context                                                                        0.1s
 => => transferring context: 2.17MB                                                                             0.1s
 => [webapp 2/5] WORKDIR /app                                                                                   0.2s
 => [webapp 3/5] COPY package.json .                                                                            0.0s
 => [webapp 4/5] RUN npm install                                                                                2.5s
 => [webapp 5/5] COPY . .                                                                                       0.1s
 => [webapp] exporting to image                                                                                 0.1s
 => => exporting layers                                                                                         0.1s
 => => writing image sha256:8287d20b5e332a528242ab3ba51f3183a18db2e0f2cb707d0c90536fdd781780                    0.0s
 => => naming to docker.io/library/bai04-webapp                                                                 0.0s
[+] Running 2/2
 ✔ Network bai04_default     Created                                                                            0.1s
 ✔ Container bai04-webapp-1  Started                                                                            0.0s

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai04>docker compose ps
NAME            IMAGE          COMMAND                          SERVICE   CREATED          STATUS          PORTS
bai04-webapp-1  bai04-webapp   "docker-entrypoint.sh node app.js"  webapp    25 seconds ago   Up 25 seconds   0.0.0.0:5000->3000/tcp
T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai04>
```

localhost:5000

localhost:5000

Chào mừng bạn đến với ứng dụng Node.js chạy trong Docker! 🚀

- Bài 5

docker-compose.yaml U

22678601_PhanTanDuy_Tuan05 > Bai05 > docker-compose.yaml

```yaml
version: "3.8"

services:
  cache:
    image: redis:alpine
    restart: always
    ports:
      - "6379:6379"
    volumes:
      - redis_data:/data

volumes:
  redis_data:
```

C:\Windows\System32\cmd.exe

```
Microsoft Windows [Version 10.0.19045.6216]
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T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai05>docker compose up -d
[+] Running 8/8
 ⠿ cache 7 layers [⠿⠿⠿⠿⠿⠿⠿]      0B/0B      Pulled                                                    9.4s
   ⠿ 589002ba0eae Pull complete                                                                        1.7s
   ⠿ cd9ebb746e5b Pull complete                                                                        1.3s
   ⠿ d3f91c0b9ad5 Pull complete                                                                        1.3s
   ⠿ cb2f87994157 Pull complete                                                                        4.7s
   ⠿ d67eeb8f9f36 Pull complete                                                                        2.5s
   ⠿ 4f4fb700ef54 Pull complete                                                                        2.9s
   ⠿ 826f751c7f6d Pull complete                                                                        3.8s
[+] Building 0.0s (0/0)                                                                   docker:default
[+] Running 3/3
 ⠿ Network bai05_default      Created                                                                  0.1s
 ⠿ Volume "bai05_redis_data"  Created                                                                  0.0s
 ⠿ Container bai05-cache-1    Started                                                                  0.2s

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai05>docker compose ps
NAME            IMAGE          COMMAND                  SERVICE   CREATED          STATUS         PORTS
bai05-cache-1   redis:alpine   "docker-entrypoint.sh redis-server"   cache     10 seconds ago   Up 8 seconds   0.0.0.0:6379->6379/tcp

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai05>
```

```
T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai05>docker compose exec cache redis-cli set myname "Phan Tan Duy"
OK

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai05>docker compose exec cache redis-cli get myname
"Phan Tan Duy"
```

- Bài 6:

docker-compose.yaml U

22678601_PhanTanDuy_Tuan05 > Bai06 > docker-compose.yaml

```yaml
version: "3.8"

services:
  db:
    image: mysql:8.0
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpassword
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppassword
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    restart: always
    depends_on:
      - db
    ports:
      - "8000:80"
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppassword
      WORDPRESS_DB_NAME: wordpress

volumes:
  db_data:
```

Select C:\Windows\System32\cmd.exe

```
T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai06>docker compose up -d
[+] Running 25/25
 ⠿ wordpress 24 layers [⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿⠿]      0B/0B      Pulled                                                    14.2s
   ⠿ ec781dee3f47 Already exists                                                                                0.0s
   ⠿ 492c65e7c8ac Already exists                                                                                0.0s
   ⠿ 60db4f438f38 Already exists                                                                                0.0s
   ⠿ 7d5440bdc2a2 Already exists                                                                                0.0s
   ⠿ f0ff10d8b131 Already exists                                                                                0.0s
   ⠿ fa79359a5906 Already exists                                                                                0.0s
   ⠿ 0e62964b059f Already exists                                                                                0.0s
   ⠿ 9ab4f7b41339 Already exists                                                                                0.0s
   ⠿ a6418fb4b014 Already exists                                                                                0.0s
   ⠿ b879b42dc415 Already exists                                                                                0.0s
   ⠿ 7ddb4115d2ee Already exists                                                                                0.0s
   ⠿ 41d15dbe17b9 Already exists                                                                                0.0s
   ⠿ 48a25387913f Already exists                                                                                0.0s
   ⠿ 5a5870c8f47c Already exists                                                                                0.0s
   ⠿ 4f4fb700ef54 Already exists                                                                                0.0s
   ⠿ 3e00916808b5 Pull complete                                                                                 4.2s
   ⠿ ce09f597207b Pull complete                                                                                 2.5s
   ⠿ 021cf943304b Pull complete                                                                                 1.2s
   ⠿ b890e1219e2a Pull complete                                                                                 2.4s
   ⠿ afc573caa3a9 Pull complete                                                                                 3.9s
   ⠿ ce8606c13c15 Pull complete                                                                                 9.4s
   ⠿ 5451487dab40 Pull complete                                                                                 5.1s
   ⠿ c61acaf42e5d Pull complete                                                                                 5.5s
   ⠿ 4413d269cb54 Pull complete                                                                                 6.3s
[+] Building 0.0s (0/0)                                                                          docker:default
[+] Running 4/4
 ⠿ Network bai06_default        Created                                                                        0.0s
 ⠿ Volume "bai06_db_data"       Created                                                                        0.0s
 ⠿ Container bai06-db-1         Started                                                                        0.1s
 ⠿ Container bai06-wordpress-1  Started                                                                        0.0s

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai06>docker compose ps
NAME                IMAGE              COMMAND                                  SERVICE     CREATED          STATUS
        PORTS
bai06-db-1          mysql:8.0          "docker-entrypoint.sh mysqld"            db          34 seconds ago   Up 33 se
conds   3306/tcp, 33060/tcp
bai06-wordpress-1   wordpress:latest   "docker-entrypoint.sh apache2-foreground"   wordpress   33 seconds ago   Up 32 se
conds   0.0.0.0:8000->80/tcp

T:\22678601_PhanTanDuy_DHKTPM18C_KT-TKPM\22678601_PhanTanDuy_Tuan05\Bai06>
```

WordPress › Installation

localhost:8000/wp-admin/install.php

English (United States)
Afrikaans
አማርኛ
Aragonés
العربية
العربية المغربية
অসমীয়া
گۆڕەی ئازربایجان
Azerbaycan dili
Беларуская мова
Български
বাংলা
বড়ো
Bosanski
Català
Cebuano
Čeština
Cymraeg
Dansk
Deutsch (Österreich)
Deutsch

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ENG 8:32 PM 4/2/2026